Yuvan Babu

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Education

Purdue University — West Lafayette, IN
Bachelor of Science in Computer Engineering — *Aug. 2026 – May 2030*

Arnold O. Beckman High School — Irvine, CA
High School Diploma; GPA 4.08 (weighted); SAT 1520; AP Scholar with Honor — *Aug. 2022 – May 2026*

Experience

Founder & Lead Hardware Builder — 2024 – Present
Chronokey — *Irvine, CA*

- Founded a custom mechanical keyboard business; delivered 15 client-specific builds to 12 paying customers, owning switch/PCB spec, assembly, firmware flashing, and final QC
- Diagnosed and repaired build failures (dead switches, shorted matrix columns, case fitment) with multimeter and soldering iron; fed root causes back into sourcing and assembly, cutting defects across builds

Applied Cybersecurity Intern — Summer 2025
Georgetown University, Applied Cybersecurity Program

- Analyzed cybersecurity risks for the Vietnam Veterans Memorial Fund as client of a Georgetown-led project
- Delivered and deployed security recommendations to the client on a one-week, full-time project timeline

Vice President & Math Instructor — 2022 – 2026
CSK Math Circle Foundation — *Irvine, CA*

- Taught weekly mathematics classes to underserved students (275+ volunteer hours); promoted to Vice President
- Redesigned the curriculum and trained volunteer instructors; 2x Gold President's Volunteer Service Award

Projects

Acheron Hospitality Smart Systems | *Provisional Patent, Inventor & System Designer* — 2024 – Present

- Designed a smart hospitality system that automates in-room settings from guest check-in
- Owned the system architecture and security logic; filed a provisional patent on the design

Custom PC Builds | *Hardware assembly, firmware, troubleshooting* — 2022 – Present

- Built three PCs from components; diagnosed POST and boot failures at the hardware level
- Debugged XMP memory instability under load: tested raised DRAM voltage in BIOS, reverted, and settled on a lower stable memory speed

IR Presence-Triggered Desktop Automation | *Arduino, IR sensor, serial, host-side scripting* — 2025

- Arduino IR break-beam sensor streams trip events over serial to a host script that fires a configurable keyboard shortcut

Independent Software Projects | *Python, Java, GitHub* — 2024 – Present

- Built and deployed a RAM-efficient Spotify web client and an algorithmic word-game solver

Leadership & Activities

Purdue Electric Racing — Aug. 2026 – Present
High Voltage Subteam Member — *West Lafayette, IN*

Founder & President, Cybersecurity Club — 2025 – 2026
Arnold O. Beckman High School — *Irvine, CA*

- Built the curriculum and ran hands-on labs teaching threat modeling and network defense

Founder & President, Machine Learning & AI Club — 2024 – 2026
Arnold O. Beckman High School — *Irvine, CA*

Technical Skills

Languages: C, C++, Python, Java, Swift
Frameworks & Tools: Git, React, Tailwind, Electron, Arduino
Hardware: Soldering and hot-swap rework, multimeter debugging, keyboard firmware (QMK/VIA), PC assembly/troubleshooting
Certifications: AWS Certified Cloud Practitioner, Java Certified Developer | **Spoken**: English, Telugu, Tamil, French